

The Surveyor v0.1 - Freeze and Forensic Baseline Report

Status: proposed functional freeze dossier for The Eye

Inspection date: 2026-06-29T22:00:19-06:00

Source branch: headless-surveyor-v0.1-dev

Frozen source commit: 01b7f6876f0af933e610eb67a3583cb628ea2ddd

Audit branch: audit/surveyor-v0.1-freeze

Executive Summary

The Surveyor v0.1 is a headless COLMAP workstation image that accepts image sets, builds one or more sparse reconstructions, selects the best candidate, normalizes that candidate to sparse/0, validates the Trainer handoff contract, and creates portable scene and evidence archives with SHA-256 records.

The source commit inspected here has a clean Git tree and passed the existing local non-GPU test suite. The latest existing GitHub Actions workflow for the same commit also completed successfully and reported a Docker Hub push. No runtime, dependency, workflow, script, test, Dockerfile, or contract change is introduced by this dossier.

The published-image digest cannot be independently verified on this Mac because Docker and Docker Buildx are unavailable. CI reported

sha256:51e9a9236cdc8c8351195709e5f0a864a01c62d4a79251f71e182219acdeb7d8; this is retained as CI-reported evidence, not promoted to an independently verified digest.

Proposed identifiers, not created by this audit:

- Git tag: surveyor-v0.1.0-smoke-validated
- Docker tag: headless-surveyor-v0.1.0

The stable Docker tag must be attached to the already-published verified digest. It must not be produced by rebuilding the image.

1. Immutable Git Identity

Field	Recorded value
Remote fetch	https://github.com/Marquezmemo/cloud-workstation.git
Remote push	https://github.com/Marquezmemo/cloud-workstation.git
Source branch	headless-surveyor-v0.1-dev
Source commit	01b7f6876f0af933e610eb67a3583cb628ea2ddd
Source tree	Clean; no staged, modified, or untracked files
git diff	Empty
git diff --stat	Empty
Existing matching Git tag	None
Audit branch	audit/surveyor-v0.1-freeze

Commands captured before creating the audit branch:

```
git switch headless-surveyor-v0.1-dev
git status --short --branch
git rev-parse HEAD
git remote -v
git diff
git diff --stat
```

Observed branch status:

```
## headless-surveyor-v0.1-dev...origin/headless-surveyor-v0.1-dev
```

2. Docker Identity

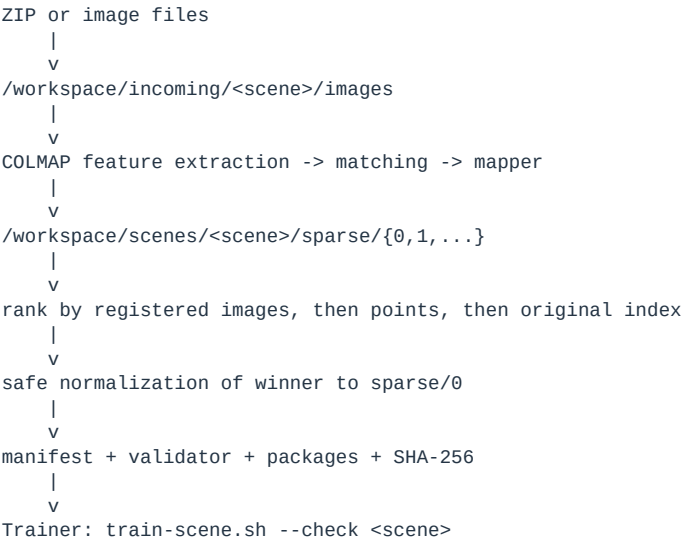
Field	Recorded value
Current development tag	docker.io/marquezmemo/cloud-workstation:headless-surveyor-v0.1-dev
Proposed stable tag	docker.io/marquezmemo/cloud-workstation:headless-surveyor-v0.1.0
Independent digest verification	PENDING
Pending reason	docker, docker image inspect, and docker buildx imagetools inspect are unavailable in the inspection environment
CI-reported pushed digest	sha256:51e9a9236cdc8c8351195709e5f0a864a01c62d4a79251f71e182219acdeb7d8
CI run	28411071377, successful, HEAD 01b7f6876f0af933e610eb67a3583cb628ea2ddd
CI job	84183939795, successful

The CI-reported digest is useful corroborating evidence but does not satisfy the requested independent pull/inspect/imagetools inspect verification. The stable tag must remain uncreated until that verification is performed from a Docker-capable host.

OCI Labels

```
org.opencontainers.image.title=cloud-workstation
org.opencontainers.image.version=headless-surveyor-v0.1-dev
org.opencontainers.image.description=Headless COLMAP image for photogrammetry reconstruction before
Gaussian Splatting training.
org.opencontainers.image.base.name=colmap/colmap:20240723.601@sha256:
73003557e3ffa36d801e71b7630c117f9d373c55f24e3afc6791b9b3d1ec01da
io.cloud-workstation.colmap.version=3.10
io.cloud-workstation.colmap.commit=0bd66d901c7549051e21e8f648777f802eb20a73
io.cloud-workstation.cuda.version=12.3.1
io.cloud-workstation.ubuntu.version=22.04
io.cloud-workstation.gdown.version=6.1.0
```

3. Runtime Architecture



The image is intentionally headless. It includes no desktop, X11, VNC, viewer, Blender, Trainer, gsp lat, or default dense-reconstruction pipeline.

The Docker entrypoint is cleared. The default command is:

```
["runpod-keepalive.sh"]
```

runpod-keepalive.sh creates the workspace roots, prints operator hints, and holds the container with sleep infinity.

4. Fixed Dependencies

Component	Frozen identity
Base image	colmap/colmap:20240723.601@sha256:73003557e3ffa36d801e71b7630c117f9d373c55f24e3afc6791b9b3d1ec01da
COLMAP	3.10 / runtime evidence 3.10-dev
COLMAP commit	0bd66d901c7549051e21e8f648777f802eb20a73
CUDA image runtime	12.3.1
Ubuntu	22.04
gdown	6.1.0, installed from a fully hashed Python lock
runpodctl	2.5.0, binary SHA-256 f484ce7d790ddc6b4a63363f3c975c70fa87bf3be1bcbad019812f6e3f4ba54e

The image also installs Bash, CA certificates, coreutils, curl, ffmpeg, findutils, gzip, jq, Python 3 and pip, rsync, SQLite, tar, and unzip. No package upgrade command is used.

5. Workspace Structure

```
/workspace/
+-- incoming/
|   +-- <scene>/
|       +-- images/
|       +-- source/<archive.zip>
+-- scenes/
|   +-- <scene>/
|       +-- images/
|       +-- database.db
|       +-- sparse/
|           | +-- 0/
|           | +-- optional additional numeric models
|           +-- surveyor-manifest.json
+-- logs/<scene>/
+-- archives/<scene>/
+-- temp/
```

sparse/0 is the normalized Trainer-ready model. Secondary sparse models remain present under their displaced numeric indices.

6. Operational Scripts And Public Commands

Script	Public command	Input	Primary output
runpod-keepalive.sh	container default	none	persistent headless process and hints
preparar-escena	preparar-escena [archivo.zip]	explicit ZIP or exactly one ZIP under /workspace	transactional incoming/<scene>/{images, source}
validate-surveyor.sh	validate-surveyor.sh	image runtime	tool/version/workspace validation
survey-scene.sh	survey-scene.sh <scene>	supported images under incoming/<scene>/images	database, sparse models, normalized sparse/0, manifest, logs
validate-surveyor-scene.sh	validate-surveyor-scene.sh <scene>	complete scene and evidence	contract result on stdout
package-surveyor-scene.sh	package-surveyor-scene.sh <scene>	validated scene and logs	scene/evidence archives and SHA-256 files

Additional operator commands include `gdown`, `runpodctl`, `sha256sum`, `colmap model_analyzer`, and Trainer-side `train-scene.sh --check <scene>` after transfer.

7. Environment Variables

Image-level

```
SURVEYOR_IMAGE_VERSION
SURVEYOR_COLMAP_VERSION
SURVEYOR_COLMAP_COMMIT
SURVEYOR_CUDA_VERSION
SURVEYOR_UBUNTU_VERSION
SURVEYOR_GDOWN_VERSION
WORKSPACE_ROOT
INCOMING_DIR
SCENES_DIR
LOGS_DIR
ARCHIVES_DIR
TEMP_DIR
```

Reconstruction

```
SCENE_NAME
INPUT_IMAGES
SCENE_PATH
LOG_DIR
MATCHER                exhaustive | sequential; default exhaustive
COLMAP_USE_GPU          1 | 0; default 1
SINGLE_CAMERA            1 | 0; default 1
CAMERA_MODEL            default OPENCV
MAX_IMAGE_SIZE          optional positive integer
OVERWRITE               true | false; default false
VALIDATOR_PATH          test/advanced override
```

Validation and packaging

```
REQUIRE_SUMMARY
ARCHIVE_DIR
ARCHIVE_PATH
EVIDENCE_PATH
TEMP_DIR
OVERWRITE
VALIDATOR_PATH
```

The frozen defaults include `CAMERA_MODEL=OPENCV` and `SINGLE_CAMERA=1`. This dossier does not recommend changing them inside v0.1.

8. Data And Processing Contracts

ZIP preparation

`preparar-escena` validates ZIP integrity with `unzip -t`, rejects unsafe paths and symbolic links, extracts to temporary storage, ignores `__MACOSX`, `.DS_Store`, and `.*`, accepts `.jpg`, `.jpeg`, `.png`, `.tif`, and `.tiff`, requires at least two non-empty images, detects case-insensitive basename collisions, and atomically publishes the incoming scene. Existing incoming or processed scenes are not overwritten.

database.db

COLMAP feature extraction creates the SQLite database. Surveyor and its validator distinguish:

- `database_images`: rows in the database image table.
- `registered_images`: images registered in the selected sparse reconstruction.

These values are not interchangeable. Validation enforces `registered_images <= database_images`.

Extraction, matching, and mapper

The frozen reconstruction sequence is:

```
colmap feature_extractor
colmap exhaustive_matcher | colmap sequential_matcher
colmap mapper
colmap model_analyzer
```

Feature extraction uses the configured camera model, single-camera policy, optional maximum image size, and GPU flag. Matching uses the selected matcher and GPU flag. The mapper may produce multiple numeric sparse-model directories.

Sparse candidate selection

Surveyor discovers canonical numeric directories under `sparse/`, requires non-empty `cameras.bin`, `images.bin`, and `points3D.bin`, and analyzes every candidate. Ranking is:

1. Highest Registered images.
2. Highest Points.
3. Lowest original numeric index for a complete tie.

If a non-zero model wins, Surveyor exchanges it with model 0 using same-filesystem temporary renames and rollback handling. The original model 0 is preserved under the winner's former index. No secondary model is deleted.

Selection evidence is written atomically to `logs/<scene>/sparse-selection.txt`. The definitive `model-analyzer.txt` is generated from normalized `sparse/0`.

Manifest

`surveyor-manifest.json` includes the schema, scene, timestamp, image/database counts, matcher, GPU flag, single-camera flag, camera model, scene path, Trainer-ready path, sparse-model count, original selected index, and registered-image count. A direct `jq -e` gate validates generated JSON before the scene validator runs.

Validation

The scene validator requires supported non-empty images, a healthy SQLite database, normalized sparse binaries, final model-analyzer metrics, selection evidence, manifest consistency, `registered_images >= 2`, `registered_images <= database_images`, candidate/model-count agreement, and GPU evidence when GPU mode was used.

Packaging and checksums

The packager first invokes the complete validator. It creates:

```
<scene>-surveyor-scene.tar.gz
<scene>-surveyor-scene.tar.gz.sha256
<scene>-surveyor-evidence.tar.gz
<scene>-surveyor-evidence.tar.gz.sha256
```

Scene archives retain normalized sparse/0 and secondary models. Evidence archives include the manifest and logs. Checksum records contain relative filenames and are verified immediately with `sha256sum -c`.

Trainer handoff

Trainer receives and extracts the scene archive, verifies its checksum, and runs:

```
train-scene.sh --check <scene>
```

Training and PLY export are outside Surveyor.

9. Publication Workflow And Tests

Workflow: `.github/workflows/publish-headless-surveyor-dev.yml`

Trigger:

- Push to `headless-surveyor-v0.1-dev`.
- Manual `workflow_dispatch`.

The workflow checks shell syntax, builds `linux/amd64`, validates the image without GPU, runs contract, sparse-selection/manifest, and ZIP tests, and pushes `headless-surveyor-v0.1-dev` only after all gates pass.

Latest existing run for the frozen source commit:

```
Run: 28411071377
Job: 84183939795
Commit: 01b7f6876f0af933e610eb67a3583cb628ea2ddd
Conclusion: success
```

No workflow was triggered by this audit branch.

10. Prueba02 Evidence

Evidence source: `docs/validation-runs/2026-06-29-prueba02-surveyor.md`, committed in the frozen source HEAD. The external evidence attachments named by that record were not present in this inspection environment and were not re-hashed here.

Evidence-backed facts:

- 30 input images.
- 30 images in `database.db`.
- Camera model `OPENCV`.
- `SINGLE_CAMERA=1`.
- Two sparse candidates.
- Original candidate 1 selected with 30 registered images.
- 4,555 points.
- 19,928 observations.
- Mean track length `4.374973`.
- Mean reprojection error `1.133537` px.
- Selected candidate normalized to `/workspace/scenes/Prueba02/sparse/0`.
- Manifest and scene validation passed.
- Scene and evidence packages were created.

- Portable checksums passed in Surveyor.
- Trainer received the scene archive and checksum, verified the checksum, extracted the scene, passed `train-scene.sh --check Prueba02`, and completed a 300-step run.

The versioned evidence record reports an evidence archive SHA-256 of `03d4f4f764e8dc4a53cff4b029781cd31877ec5c795db9f39b6e962dfed21157` and size of 5,802 bytes. This audit records that statement but does not claim a new local verification because the archive was unavailable.

Esta evidencia valida funcionalidad, reconstrucción sparse y handoff. No valida calidad profesional ni comercial.

11. Non-Destructive Tests Executed By This Audit

Environment:

```
Date interval: 2026-06-29T22:03:31-06:00 to 2026-06-29T22:03:54-06:00
Host: macOS 26.5.1 (25F80), Darwin 25.5.0, ARM64
Bash: 3.2.57
Python: 3.14.6
jq: 1.8.2
SQLite: 3.51.0
GPU: not used
Docker: unavailable
```

Command	Exit	Result
<code>bash -n scripts/*.sh scripts/preparar-escena tests/*.sh</code>	0	Shell syntax passed; no output
<code>tests/test-surveyor-contract.sh</code>	0	Valid scene, packages, GPU evidence fixture, and negative contract cases passed
<code>tests/test-survey-scene-manifest.sh</code>	0	Registered-image selection, point tiebreak, keep-zero, validation, and packaging passed
<code>tests/test-preparar-escena.sh</code>	0	ZIP/macOS metadata, autodetection, corruption, collision, existing-scene, and unsafe-path cases passed

No GPU run, image build, Docker pull, Docker tag, Docker push, or new publication workflow was executed.

12. Limitations

- No professional or commercial quality threshold is defined.
- OPENCV distortion versus a PINHOLE/undistorted workflow remains unevaluated.
- Trainer-side archive extraction remains manual.
- Dense reconstruction and PLY export are outside Surveyor v0.1.
- Video frame extraction is not implemented.
- The stable image digest still requires independent verification with Docker-capable tooling.
- The Prueba02 attachments were not locally present for independent re-verification.

13. Accidental Reconstruction And Publication Risks

- Any commit pushed to `headless-surveyor-v0.1-dev` triggers a build and overwrites the mutable development tag.
- Rebuilding from the same Git commit can produce a different image digest because external build tooling, timestamps, or registry metadata may differ.
- Creating `headless-surveyor-v0.1.0` through a new build would violate the freeze requirement.
- Moving the proposed Git tag after creation would destroy source immutability.

- Retagging before independently verifying the existing digest could freeze the wrong artifact.
- Editing official documentation from the audit branch would mix evidence preparation with canonical documentation ownership.

14. Recovery Procedure

1. Confirm the approved Git tag points exactly to 01b7f6876f0af933e610eb67a3583cb628ea2ddd.
2. From a Docker-capable host, inspect the current development tag without rebuilding:


```
docker buildx imagetools inspect \
  docker.io/marquezmemo/cloud-workstation:headless-surveyor-v0.1-dev
```
3. Confirm the returned digest against the CI-reported digest and record the independently verified value.
4. Create headless-surveyor-v0.1.0 by registry retag/copy of that exact digest. Do not run `docker build`.
5. Pull the stable tag and confirm the same digest with `docker image inspect` or `docker buildx imagetools inspect`.
6. Run `validate-surveyor.sh` in the pulled image. A new GPU reconstruction is not required for restoration if Prueba02 evidence remains accepted.
7. Restore scenes from verified scene/evidence archives and run their `.sha256` checks before Trainer handoff.

The exact registry-retag command depends on the approved registry tooling and credentials. The Eye should document the authorized operational command after digest verification.

15. Handoff To The Eye

The Eye should integrate:

1. Frozen Git identity and clean-tree evidence.
2. Independently verified Docker digest once obtained.
3. Approval and creation status for the proposed Git and Docker tags.
4. Runtime architecture, fixed dependency table, workspace contract, and public commands.
5. Distinction between `database_images` and `registered_images`.
6. Sparse ranking and normalization contract.
7. Manifest, validator, package, checksum, and Trainer handoff contracts.
8. Prueba02 facts and the explicit non-commercial-quality disclaimer.
9. Local test record and latest successful CI record.
10. Recovery procedure and warning against rebuilding or pushing to the development branch.

16. Freeze Declaration

This audit branch adds only proposal documents and generated forensic artifacts. It makes no functional change to The Surveyor v0.1. The source commit remains 01b7f6876f0af933e610eb67a3583cb628ea2ddd; runtime files, workflows, dependencies, contracts, and behavior remain untouched.